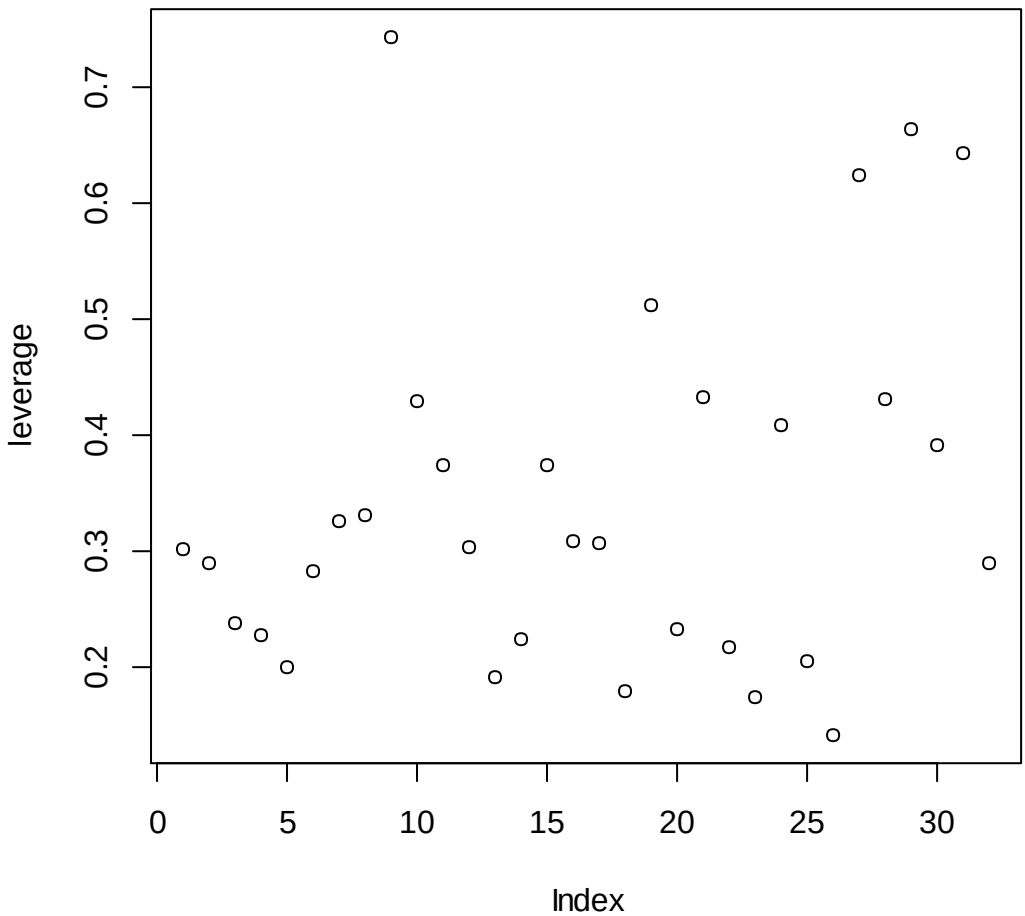
R Lab 12. Influential Data Analysis

```
> reg = lm(mpg~.,data=mtcars)
> infl = influence(reg)
> names(infl)
[1] "hat"         "coefficients" "sigma"       "wt.res"
```

 # Contains hat diagonals H_{ii}, vector of coefficients (without the ith case), s = RMSE (without the ith case)

```
> leverage = infl$hat
> p = 10; n = length(mpg); mean_leverage = (p+1)/n;
> mean_leverage
[1] 0.34375
> mean(leverage)
[1] 0.34375
> plot(leverage)
```




```
> infl$coefficients
               (Intercept)          cyl         disp           hp
Mazda RX4        -1.51953212  0.0066222621 -1.746129e-03  5.728104e-03
Mazda RX4 Wag     0.09977168 -0.0343756932 -1.779194e-04  3.133630e-03
Datsun 710       -4.49406953  0.1397388420  5.180829e-03 -6.441502e-03
  < truncated >
```

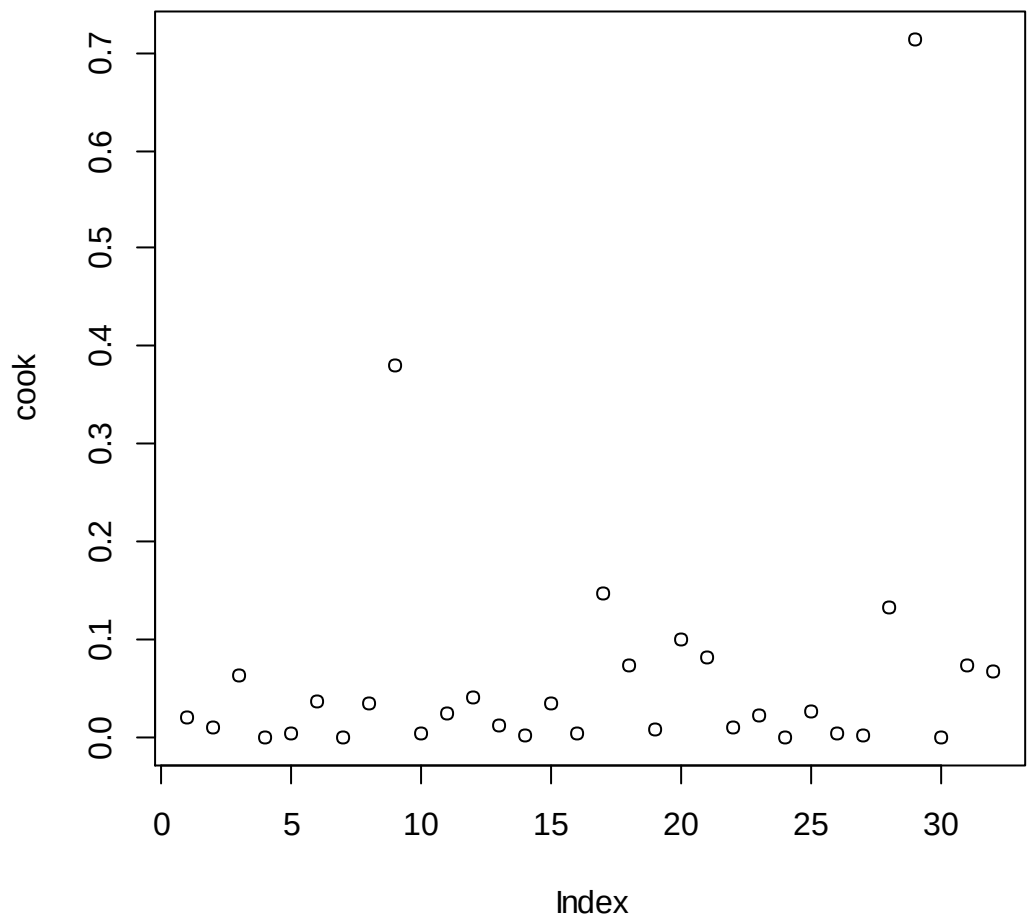
```
> infl$sigma
       Mazda RX4      Mazda RX4 Wag         Datsun 710      Hornet 4 Drive
        2.681664          2.699561           2.567603           2.715329
   < truncated >
```

 # ADDITIONAL PACKAGE "CAR" (Go to "Packages" tab and choose "car")

```
> install.packages("car")
> library(car)
> outlierTest(reg)

   rstudent unadjusted p-value Bonferonni p

321 4.272284        2.4397e-05    0.0095635
```

 # Cook's Distance

```
> cook = cooks.distance(reg)
> cooks.distance(reg)
       Mazda RX4      Mazda RX4 Wag         Datsun 710      Hornet 4 Drive
     2.059098e-02       9.218355e-03       6.352411e-02       1.306707e-04

> plot(cook)
```



 # The Cook's distance measures the effect of deleting the i-th observation

> cook[cook > 0.3]
Merc 230 Ford Pantera L
0.3792206 0.7135206

> influence.measures(reg)
 # Besides the Cook's distance, it calculates DFBETS, DFFITS, and other measures of influence

```
> influence.measures(reg)
Influence measures of
         lm(formula = mpg ~ ., data = mtcars) :

                      dfb.1_   dfb.cyl  dfb.disp    dfb.hp  dfb.drat    dfb.wt  dfb.qsec
dfb.vs    dfb.am  dfb.gear  dfb.carb    dffit cov.r   cook.d   hat inf
Mazda RX4           -0.080228  0.006263 -0.096634  0.260049  0.035485  0.109069 -0.015734
0.098621 -0.139736  0.166814 -0.258558 -0.47034 1.859 2.06e-02 0.303
Mazda RX4 Wag        0.005233 -0.032293 -0.009781  0.141320  0.003009  0.013676 -0.046659
0.090681 -0.129349  0.086761 -0.126009 -0.31261 2.115 9.22e-03 0.290
Datsun 710          -0.247818  0.138020  0.299453 -0.305427  0.187003 -0.385911  0.273358
-0.190640 -0.371231  0.102704  0.406685 -0.86281 0.655 6.35e-02 0.239
```